College Board AP® Chemistry

Your notes

Intermolecular & Interparticle Forces

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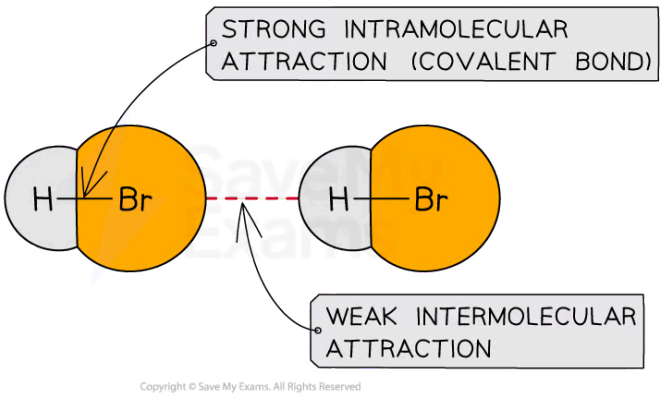
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What Are Intermolecular Forces?

- Intermolecular forces are weak forces that exist between electrically neutral molecules
- All intermolecular interactions are electrostatic
  - This means that they involve attractions between positive and negative species, much like ionic bonds
- Intermolecular forces are generally **weaker** than **intramolecular attraction forces** like ionic bonds, covalent bonds and metallic bonds
- Intermolecular forces are responsible for the physical properties of **covalent molecules** including:
  - Boiling point
  - Melting point
  - Density
  - Viscosity
  - Solubility
- For example, the **stronger the intermolecular force**, the **higher the boiling and melting point** of the molecule

Intermolecular Forces Vs Intramolecular Forces



Weak intermolecular forces exist between molecules while stronger intramolecular forces exist between the atoms that make up the molecules

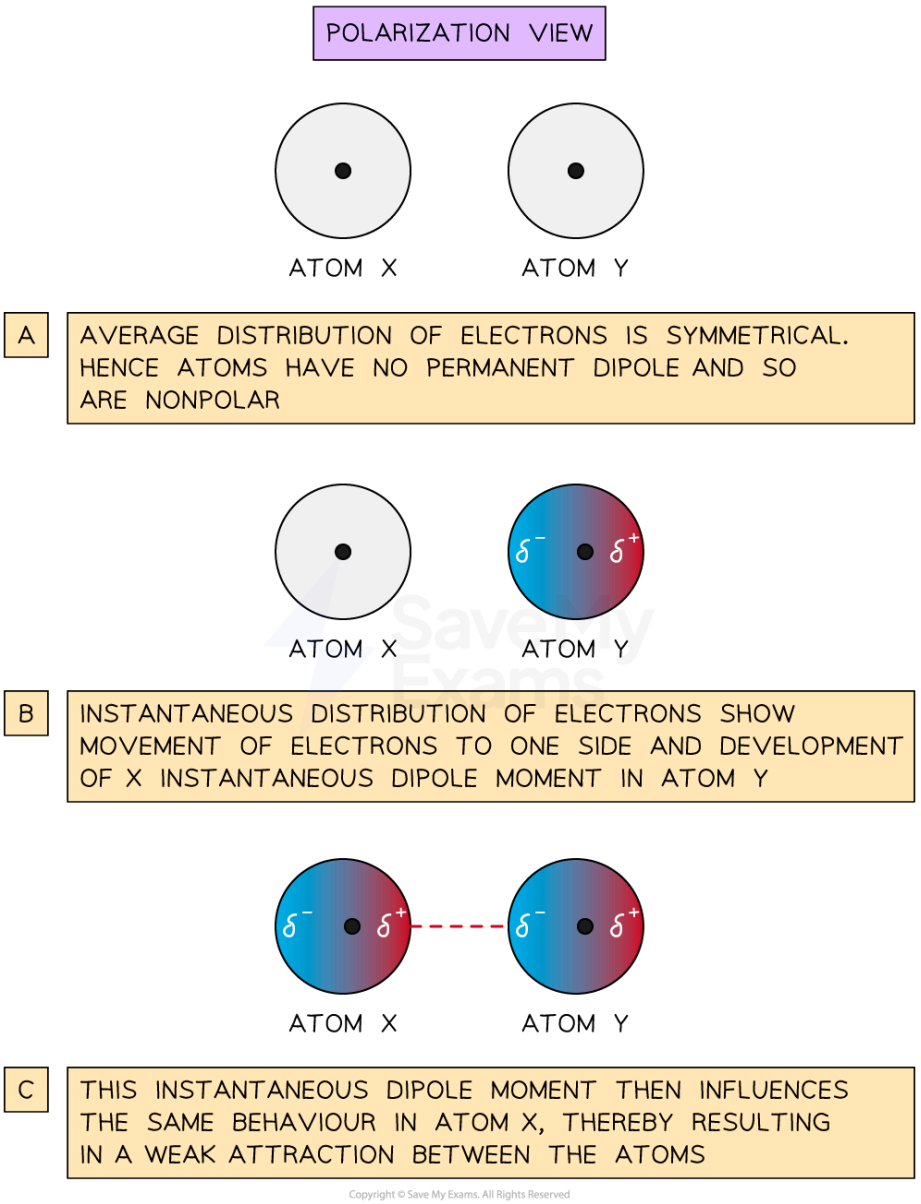
London Dispersion Forces

- London dispersion forces are a weak type of intermolecular forces produced by *small* ***instantaneous dipoles*** that occur in **nonpolar** molecules

- Instantaneous dipoles are the result of an electron's motion in one atom affecting another atom
- The **average distribution** of electrons in a nonpolar molecule is symmetrical
  - This means that they lack a permanent dipole moment
- However, at any instant due to their motion, these electrons become unevenly distributed
  - This means that one side develops a partial negative **charge, δ–** and the other side develops a partial positive **charge, δ+**
- The instantaneous dipole that develops in one molecule then influences another molecule resulting in a weak attraction

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London Dispersion Forces



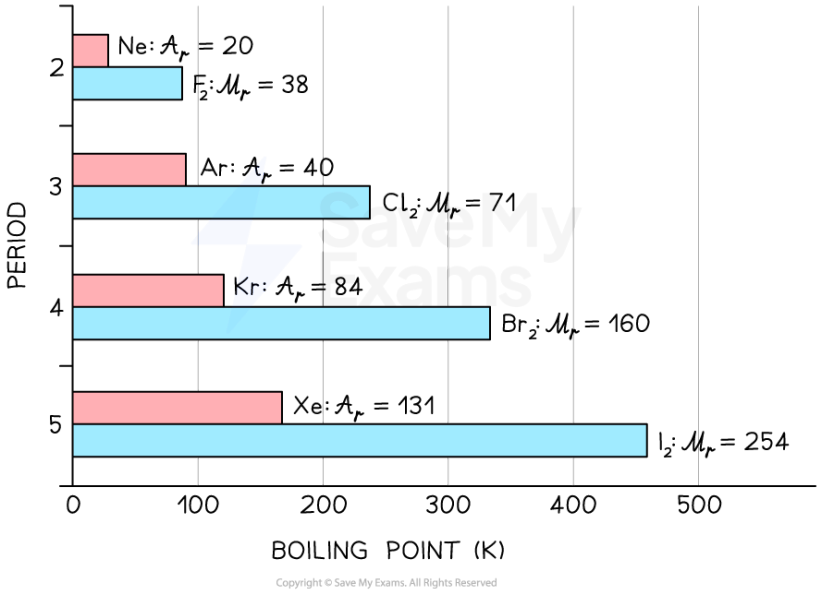
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A snapshot of the average and instantaneous distribution of electrons in an atom to demonstrate the development of London dispersion forces.

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- The strength of London forces is influenced by **polarizability**
- Polarizability describes the ease of charge distortion in a molecule and the consequent development of an instantaneous dipole
- The greater the polarizability, the more easily the electron cloud can be distorted to give an instantaneous dipole
  - This results in a stronger dispersion force
- Polarizability increases with **increasing number of electrons** in a molecule
  - Hence, the strength of dispersion forces increases with **increasing atomic or molecular size**
- For example, the boiling points of halogens and noble gases increase as you go down the group, where dispersion forces are the only forces responsible for this property

Boiling Points of Halogens & Noble Gases



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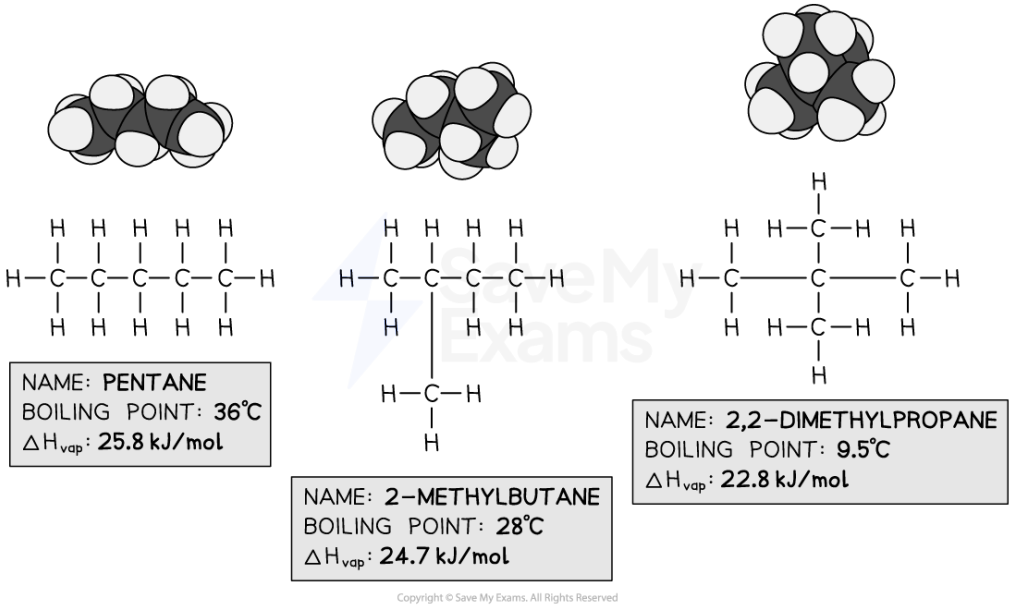
The boiling point of halogens and noble gases increases with the number of electrons

- For molecules of about the same molecular mass, polarizability is influenced by branching/compactness
  - The more branched the molecule, the less polarizable the molecule is and the smaller its dispersion force
- For example, the compounds **pentane, 2-methylbutane, and 2,2-dimethylpropane** all have the same molecular formula, C_5H_{12}, and same molecular mass, 72.15 g mol^{-1}
  - However, they differ in the arrangement of their atoms

- Pentane has no branching and is the least compact
- 2-methylbutane has one branch making it more compact than pentane
- 2,2-dimethylpropane has two branches making it more compact than 2-methylbutane
- This increase in the branching/compactness results in a decrease in boiling point as you move from pentane to 2-methylbutane to 2,2-dimethylbutane

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Branching & Polarizability



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The effect of branching on the strength of dispersion forces for molecules of the same molecular size.

Examiner Tips and Tricks

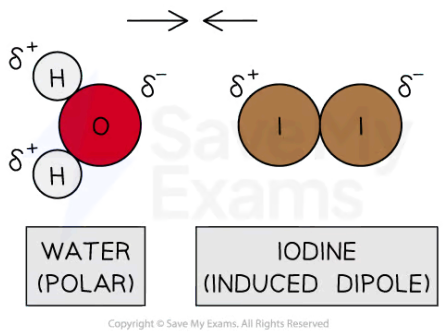
Dispersion forces are typically present in all covalent molecules but are the only forces present in nonpolar molecules like hydrocarbons.

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Dipole-Induced Dipole Interactions

- This is a weak attraction that results from the interaction between a **polar molecule** and a nonpolar molecule
- The polar molecule induces, or causes, a temporary dipole in the nonpolar molecule by distorting its electron distribution
- For example, in the interaction between polar water molecules and nonpolar iodine molecules:
  - There is a partial negative charge on the oxygen atom of the polar **O—H** bond in water
  - This causes a distortion in the electron cloud distribution in the iodine molecules
  - This distortion creates an induced dipole in the iodine molecule
- This induced dipole is then weakly attracted to the negative oxygen atom, allowing some degree of iodine dissolution in water
- However, because this force is weak, the iodine molecules are not very soluble in water
  - If the water becomes supersaturated with iodine, then there are not enough water molecules to cause this attraction

Dipole-induced dipole interactions



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Dipole-induced dipole interaction between polar water molecules and iodine molecules

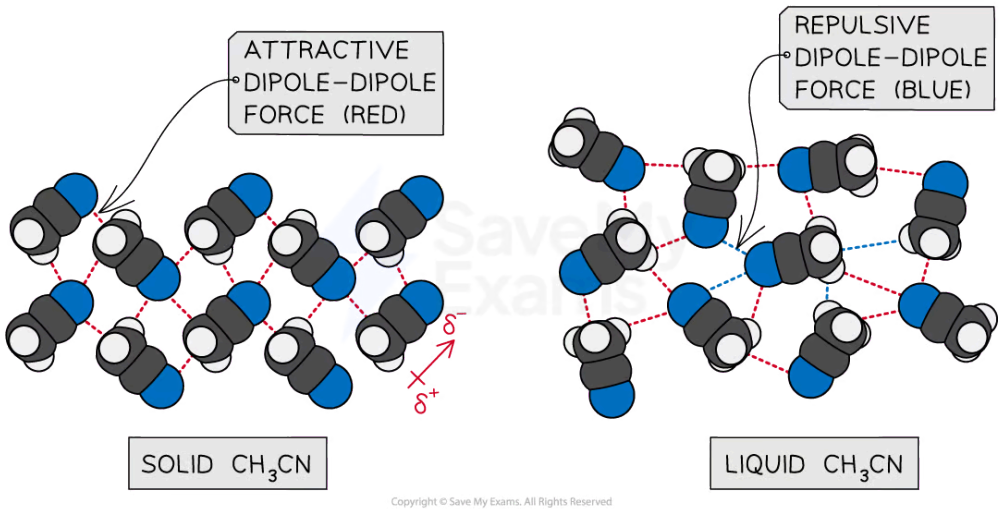
- The strength of this interaction is dependent on two factors:
  - The magnitude of the dipole moment in the polar molecule
  - The polarizability of the nonpolar molecule

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Dipole-Dipole Interactions

- Dipole-dipole interactions are a consequence of permanent dipole moments in polar molecules
- The dipole-dipole force is an attractive intermolecular force
  - This is because polar molecules tend to arrange themselves so that the positive end of one molecule is near the negative end of another
- Repulsions can also occur when the positive (or negative) ends of two molecules are in close proximity
- To illustrate this force, consider the way the molecules of acetonitrile, **CH_3CN** are packed in the liquid and solid states
  - In the solid state, they are arranged with the negatively charged nitrogen end of the molecule aligned with the positively charged $—CH_3$ end
  - In the liquid state, however, as a result of less restriction in the movement of molecules, the arrangement is disordered
    - This means both attractive and repulsive forces are present at any instant

Dipole-Dipole Interactions



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Dipole-dipole interactions between molecules of acetonitrile, CH_3N in solid and liquid states.

- We can see the effect of dipole-dipole interactions in polar molecules by comparing the boiling points of acetonitrile and propane, C_3H_8
  - Acetonitrile is a polar molecule with a **molecular weight of 41 amu** that boils at **355 K**

- Propane is a nonpolar molecule with a **molecular weight of 44 amu** that boils at **213 K**
- Dispersion forces are present in both molecules
  - Since both molecules have a similar molecular weight, the strength of dispersion forces will be similar
- However, the additional presence of a dipole-dipole force in acetonitrile gives it a higher boiling point
- Generally, the strength of dipole-dipole interaction is dependent on the magnitude of the dipole and the proximity of the molecules



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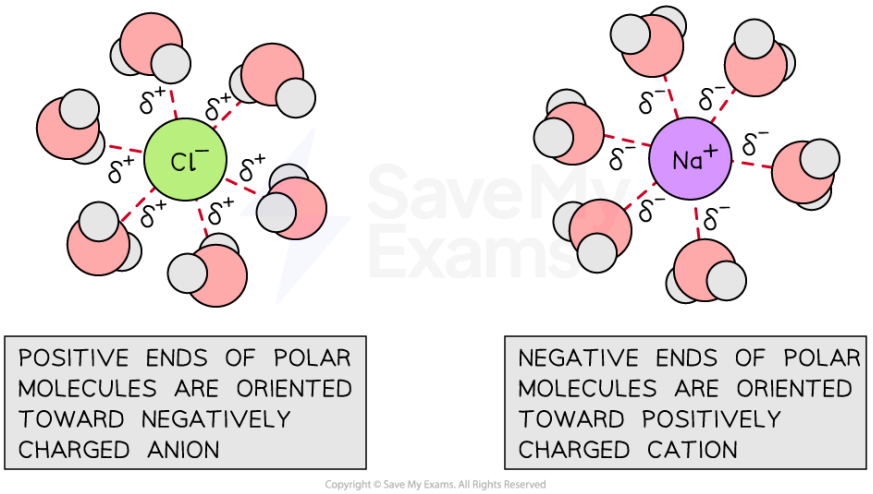
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Ion-Dipole Interactions

- This force which exists between an ion and a polar molecule, accounts for the dissolution of ionic compounds in polar solvents like water
- For example, in a solution of sodium chloride, NaCl, in water:
  - The oxygen atom has a partial negative (δ^-) charge
  - The hydrogen atoms have a partial positive charge (δ^+)
  - Overall, there is a dipole moment within the water molecule
    - The Na^+ ion is attracted to the negative end of the dipole
    - The Cl^- ion is attracted to the positive end of the dipole

Ion-Dipole Interactions



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Ion-dipole interaction between sodium and chloride ions with water

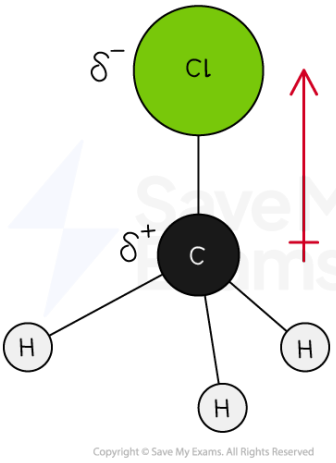
- The magnitude of ion-dipole interactions increases as either the ionic charge or the magnitude of the dipole moment increases

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Molecular Dipole Moment

- Molecular dipole moments give information about the separation of charges in a molecule
- The dipole moment (?) is a measure of the net polarity in a molecule and is measured in **Debyes (D)**
  - It is a product of the charge, Q, at the end of a dipole and the distance, *r*, between the charge
- The greater the difference in electronegativity between the bonded atoms of the molecule, the greater the dipole moment of that molecule
- For example, in chloromethane, the **C—Cl** bond has a dipole moment because chlorine is more electronegative than carbon
  - The chlorine atom develops a partial negative (δ^-) charge
  - The carbon atom develops a partial positive charge (δ^+)
  - Therefore, the **C—Cl** bond is described as a polar bond
- The dipole moment in the molecule is represented using an arrow with a plus at the end
  - The plus end of the arrow represents the positive atom or end of the polar bond / dipole moment
  - The arrowhead represents the negative atom or end of the polar bond / dipole moment

Molecular Dipole Moment



A dipole moment exists between C-Cl bonds due to the large difference in electronegativity between the carbon and chlorine atoms.

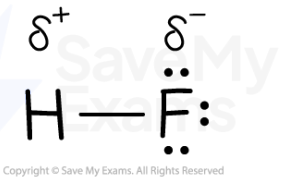
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Hydrogen Bonding

- This is a special type of dipole-dipole interaction found in covalent molecules in which hydrogen is directly bonded to either a **nitrogen, oxygen** or **fluorine** atom
  - These elements are very electronegative
  - Therefore, they form polar bonds with hydrogen

Polar covalent bond H—F bond

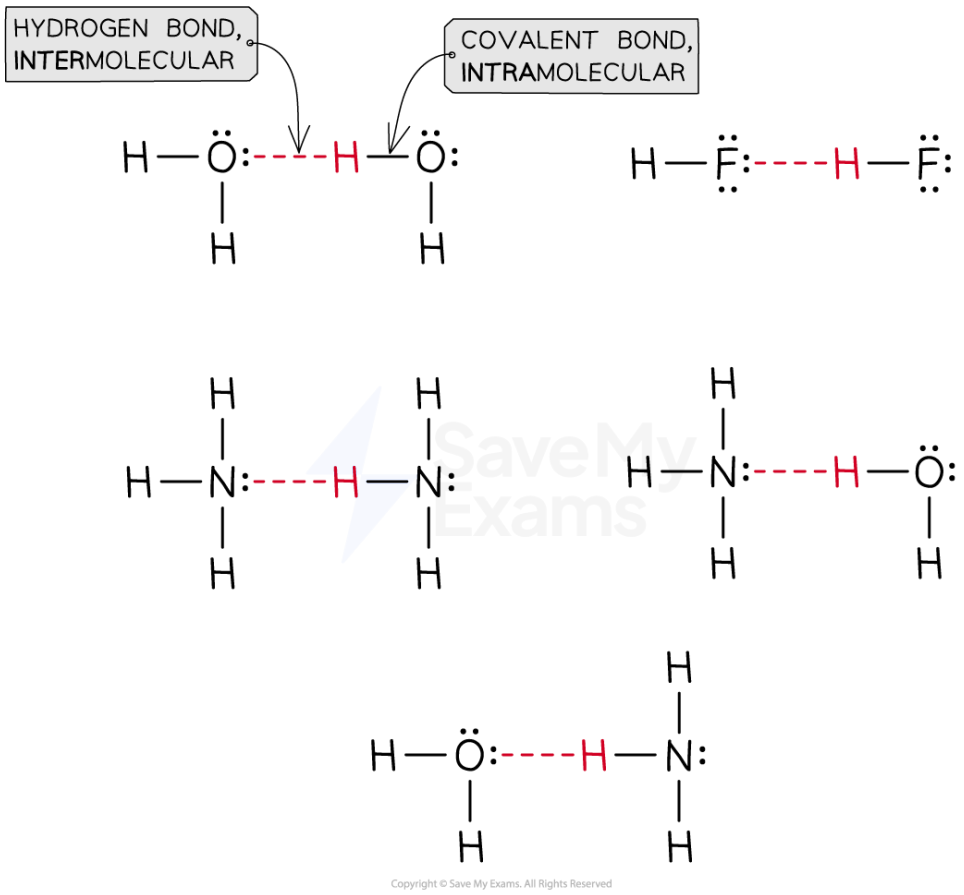


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A polar covalent bond in hydrogen fluoride is described by the presence of δ+ on the hydrogen and δ- charges on the electronegative fluorine atom

- The hydrogen atom is significantly less electronegative than a **nitrogen, oxygen** or **fluorine** atom
  - Therefore, the hydrogen atom has a partial positive (δ+) charge
  - The very electronegative element has a partial negative charge (δ-)
- **Hydrogen bonding** describes the attraction between the δ+ hydrogen atom ($H^{\delta+}$) in this polar bond and the nonbonding electron pairs on a nearby small electronegative atom or ion
  - Compared to dipole-dipole or London dispersion forces, hydrogen bonds are a stronger intermolecular force

Hydrogen Bonding

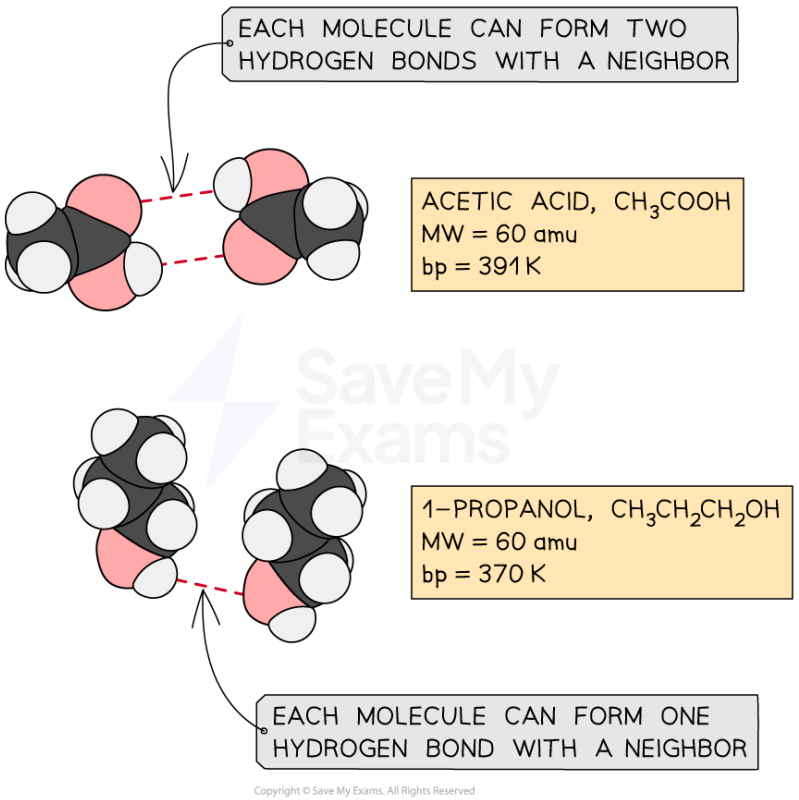


Hydrogen bonding between molecules of ammonia, water and hydrogen fluoride

- The strength of the hydrogen bond is highest in molecules where the hydrogen atom is bonded to a fluorine atom
  - This is because fluorine has the highest electronegativity
- Generally, the strength of hydrogen bonds increases in the order:
  H—N < H—O < H—F
- However, the strength of hydrogen bond in a molecule is also dependent on how many hydrogen bonds can be formed per pair of molecules
- For example, acetic acid, CH_3COOH, and propanol, $CH_3CH_2CH_2OH$:
  - They both have the same molecular mass, 60 amu
  - They are both capable of forming hydrogen bonds
- However, a pair of acetic acid molecules can form two hydrogen bonds, whereas a pair of propanol molecules can form only one
  - Hence, the boiling point of acetic acid is greater than propanol

Multiple Hydrogen Bonds

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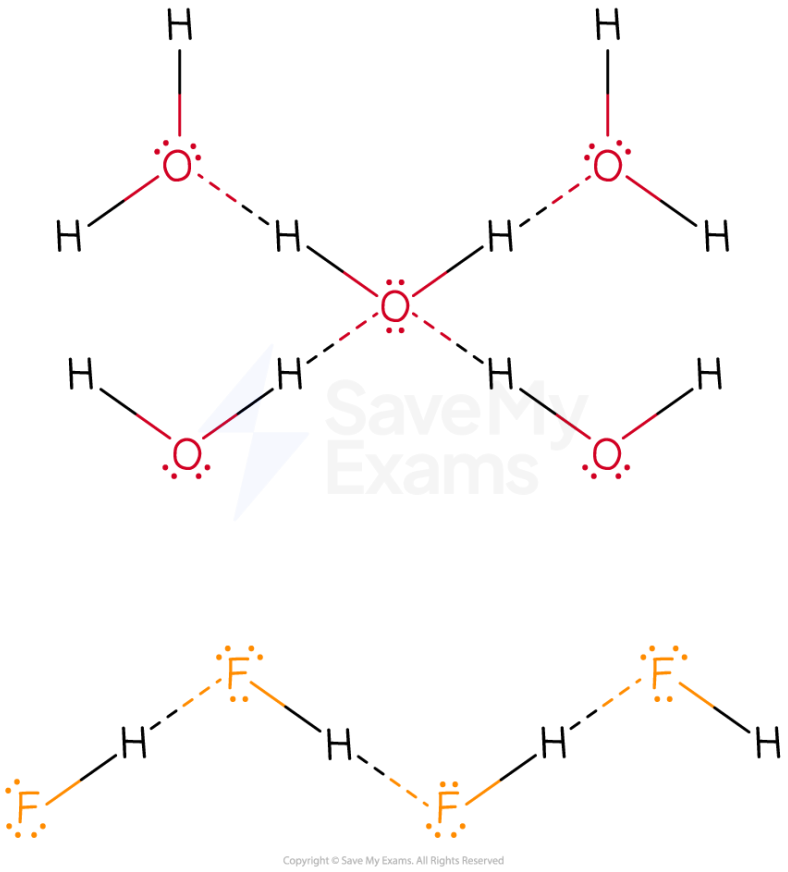
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Hydrogen bonding in acetic acid and propanol. Each acetic acid can form two hydrogen bonds while propanol can form only one hydrogen bond.

- Similarly, the hydrogen bonds in water are greater than that in hydrogen fluoride.
  - Water can form two hydrogen bonds
    - This is because there are 2 lone pairs of electrons on the oxygen atom AND two available δ+ hydrogen atoms
  - Hydrogen fluoride forms only one hydrogen bond
    - This is because, although there are 3 lone pairs of electrons on the fluorine atom, there is only one available δ+ hydrogen atom
  - Hence water has a higher boiling point than hydrogen fluoride

Hydrogen Bonding in H_2O and HF

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The diagram shows two hydrogen bonds per pair of water molecules compared to only one per pair of hydrogen fluoride

Examiner Tips and Tricks

- A covalent molecule may not be able to form hydrogen bonds with itself due to the lack of a polar hydrogen atom yet forms hydrogen bonds with another molecule with a polar hydrogen atom
- For example, acetaldehyde, CH_3CHO, can form hydrogen bonds with water molecules which makes it soluble in water

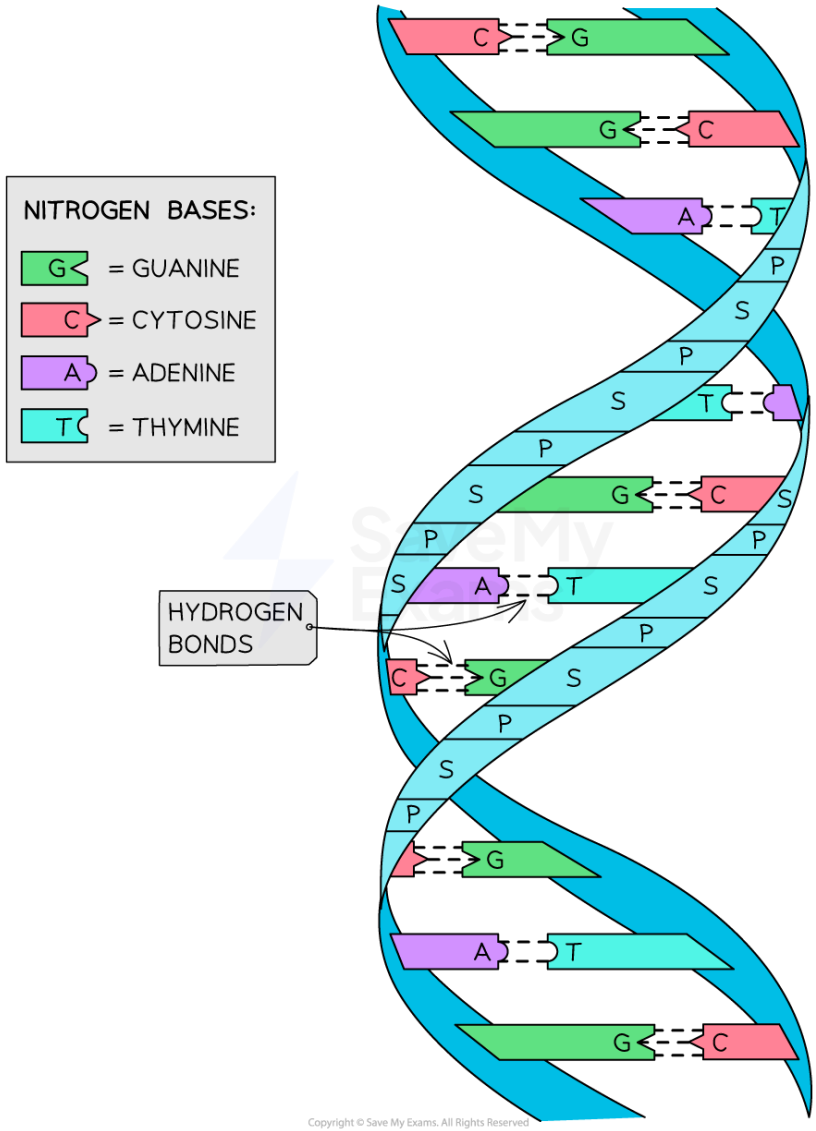
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Interactions in Large Biomolecules

- Intermolecular interactions are also found in large biomolecules like **proteins, carbohydrates and lipids**
- These interactions are possible either between biomolecules or within different regions of the same molecule
- For example, double-helix **DNA molecules** are held together by hydrogen bonds
  - The hydrogen bonding involves the nitrogenous base components between each DNA molecule strand

Hydrogen Bond in DNA Strands

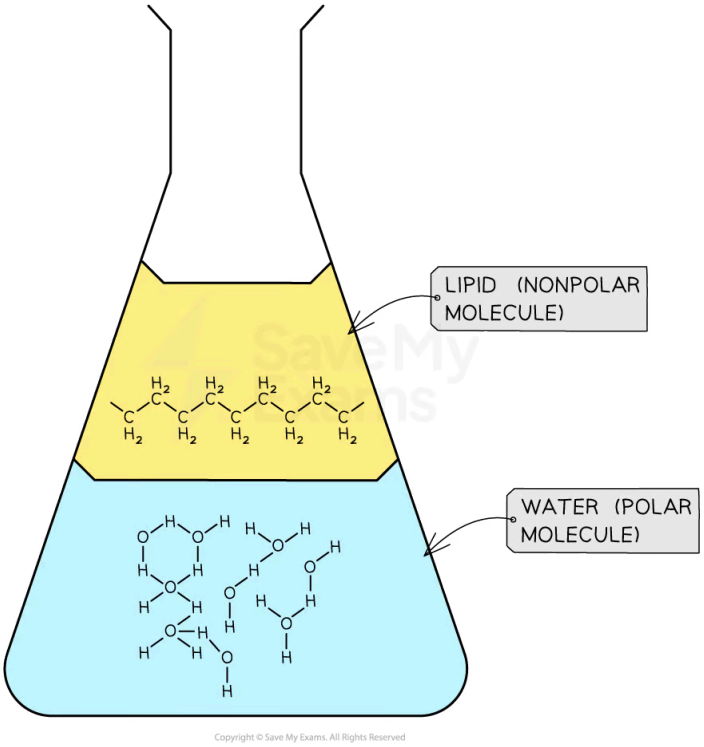


Double helix DNA strands show the hydrogen bonds between the nitrogenous base components of each strand

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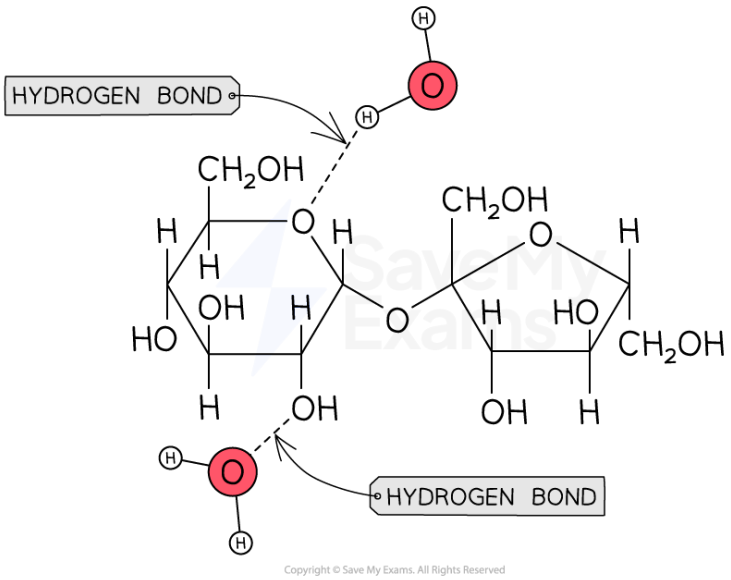
- Interaction between different biomolecules is seen between:
  - The **hydrophobic — "water fearing" —** tails of lipids and water molecules, which makes lipids insoluble in water due to the lack of hydrogen bond
  - The **hydrophilic — "water loving" —** interaction between carbohydrates and water molecules, which makes them soluble in water

Lipids Interaction with Water



Nonpolar oil molecules do not mix with water due to a lack of polar hydrogen

Carbohydrate Interaction with Water



The presence of polar hydrogen atoms on sugar molecules makes glucose dissolve in water

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